🔶 SECTION 3: Imperfections in Solids (Q81–110)

81. Schottky defect in NaCl decreases density by:
    A) Slightly
    B) Significantly
    C) Does not change
    D) Doubles
82. Frenkel defect is common in:
    A) AgCl
    B) NaCl
    C) KCl
    D) CsCl
83. Colour of NaCl changes due to:
    A) F-centres
    B) Schottky defect
    C) Frenkel defect
    D) None
84. Vacancy defect increases with:
    A) Temperature
    B) Pressure
    C) Magnetic field
    D) Voltage
85. Fraction of missing atoms in Schottky defect = 10^{-6}. Density change = ?
    A) Negligible
    B) Significant
    C) 50%
    D) Zero
86. Edge dislocation affects:
    A) Mechanical properties
    B) Electrical properties
    C) Magnetic properties
    D) Optical properties
87. Screw dislocation is:
    A) Line defect
    B) Surface defect
    C) Point defect
    D) Volume defect
88. F-centres result from:
    A) Electron trapped in anion vacancy
    B) Missing cation
    C) Extra interstitial atom
    D) None
89. Schottky defect increases with:
    A) Temperature
    B) Magnetic field
    C) Pressure
    D) Voltage

90. Frenkel defect increases with:
A) Temperature
B) Pressure
C) Magnetic field
D) Voltage

91. In NaCl, fraction of Schottky defects = 10^{-6}. Number of missing ions per mole ≈ ?
A) 6×10^{17}
B) 6×10^{18}
C) 6×10^{19}
D) 6×10^{20}

92. Number of vacancies in NaCl at 1000 K, given formation energy = 2 eV, k = 8.617×10^{-5} eV/K. Use n = N e^(-E/kT). Approx. n/N = ?
A) 10^{-10}
B) 10^{-8}
C) 10^{-6}
D) 10^{-4}

93. Colour centres in KCl produce:
A) Blue colour
B) Green colour
C) Red colour
D) Yellow colour

94. Frenkel defect does not affect:
A) Density
B) Electrical conductivity
C) Diffusion
D) Both B & C

95. Schottky defect affects:
A) Density
B) Conductivity
C) Mechanical strength
D) Both A & B

96. Edge dislocation line along x-axis. Burgers vector perpendicular to:
A) x-axis
B) y-axis
C) z-axis
D) None

97. In ionic crystals, F-centres are responsible for:
A) Colour
B) Hardness
C) Density
D) Thermal conductivity

98. Fraction of Frenkel defects increases with:
A) Temperature
B) Pressure
C) Impurity
D) None

99. Grain boundary defect is:
A) Line defect

B) Surface defect
C) Point defect
D) Bulk defect

100. Stacking fault occurs due to:
A) Misalignment in planes
B) Missing atoms
C) Interstitial atoms
D) Substitutional impurity

101. Number of vacancies in Schottky defect at T = 1200 K, formation energy = 1.2 eV, Boltzmann constant k = 8.617×10^{-5} eV/K ≈ ?
A) 10^{-4} fraction
B) 10^{-6} fraction
C) 10^{-8} fraction
D) 10^{-10} fraction

102. Edge dislocation increases:
A) Strength
B) Conductivity
C) Magnetic property
D) Density

103. Frenkel defect is common in:
A) AgBr
B) NaCl
C) KCl
D) CsCl

104. Schottky defect common in:
A) NaCl, KCl
B) AgBr
C) ZnS
D) CsCl

105. F-centres form when:
A) Electron trapped in anion vacancy
B) Cation missing
C) Extra atom present
D) None

106. Thermal activation increases:
A) Defect concentration
B) Density
C) Mass
D) None

107. Frenkel defect is also called:
A) Displacement defect
B) Vacancy defect
C) Interstitial defect
D) Colour centre

108. Schottky defect occurs in:
A) NaCl, KCl
B) AgCl

C) ZnS
D) CsCl

109. Fraction of missing atoms in Schottky defect = 10^{-5}. Effect on density = ?
A) Slight decrease
B) Significant decrease
C) No change
D) Doubled

110. Edge dislocation line movement causes:
A) Plastic deformation
B) Thermal expansion
C) Electrical conductivity
D) Colour change

🔶 SECTION 4: Electrical & Magnetic Properties (Q111–130)

111. Conductivity of intrinsic semiconductor increases with:
A) Temperature
B) Pressure
C) Magnetic field
D) Voltage

112. Conductivity of metal decreases with:
A) Temperature increase
B) Pressure
C) Magnetic field
D) Voltage

113. N-type semiconductor contains:
A) Extra electrons
B) Extra holes
C) Neutral atoms
D) Ions

114. P-type semiconductor contains:
A) Extra holes
B) Extra electrons
C) Neutral
D) Ions

115. Diamagnetic material is:
A) Weakly repelled by magnetic field
B) Weakly attracted
C) Strongly attracted
D) None

116. Paramagnetic material is:
A) Weakly attracted
B) Weakly repelled
C) Strongly attracted
D) Neutral

117. Ferromagnetic materials have:
A) Strong attraction

B) Weak attraction
C) Repelled
D) Neutral

118. Curie temperature is:
A) Ferromagnetic → paramagnetic transition
B) Melting point
C) Boiling point
D) Freezing point

119. Superconductors have:
A) Zero resistance
B) High resistance
C) Medium resistance
D) Infinite resistance

120. Meissner effect:
A) Expulsion of magnetic field
B) Attraction
C) Heating
D) Cooling

121. Hall effect used to determine:
A) Type of charge carriers
B) Resistivity
C) Current
D) Voltage

122. Piezoelectric solids generate:
A) Voltage on stress
B) Heat
C) Magnetic field
D) None

123. Examples of piezoelectric solids:
A) Quartz, Rochelle salt
B) NaCl
C) Diamond
D) Al

124. Magnetic domains are:
A) Regions of aligned magnetic moments
B) Crystals
C) Defects
D) None

125. Paramagnetic susceptibility:
A) Positive
B) Negative
C) Zero
D) Infinite

126. Diamagnetic susceptibility:
A) Negative
B) Positive
C) Zero
D) Infinite

127. Superconductivity destroyed above:
    A) Critical temperature
    B) Melting point
    C) Boiling point
    D) Curie temperature
128. Conductivity of extrinsic semiconductor increases with:
    A) Doping
    B) Temperature
    C) Both
    D) None
129. Example of ferromagnetic material:
    A) Fe
    B) Al
    C) Cu
    D) Na
130. Example of diamagnetic material:
    A) Cu
    B) Fe
    C) Ni
    D) Co